AHMED GHABEN

Moscow, Russia | ghabenahmed25@gmail.com | ahmedghaben.github.io | github.com/AhmedGhaben | linkedin.com/in/ahmedghaben

Applied Mathematics & Computer Science | Software Engineering | Quantitative Research

EDUCATION

Moscow Institute of Physics and Technology (MIPT), Moscow, Russia Expected 2028

B.Sc. Applied Mathematics and Computer Science, English-taught program | GPA: 9.14/10 (3.92/4.0)

RESEARCH EXPERIENCE

Combinatorial Optimization Research, MIPT 2026

Supervised by Danil Sagunov | LIPS 2026 Summer Research Program; presented at the LIPS conference

Diverse Matroid Bases and Spanning Trees Manuscript in preparation

- Conducted algorithms research in combinatorial optimization; proved polynomial-time solvability for three diverse matroid bases using O(r) oracle calls and obtained O(n) time for spanning trees.
- Extended the result to four diverse spanning trees and established the corresponding four-basis result for general matroids.

Diverse Maximum Cuts in Planar Graphs Manuscript in preparation

- Characterized when a graph admits three maximum cuts at the largest possible pairwise distance, reducing a search over triples to a search over pairs.
- Derived a polynomial-time decision procedure for planar graphs and computationally verified the characterization on 23,000+ graph instances.

SELECTED PROJECTS

Cloud Point-of-Sale System | Founder In development

TypeScript, React, NestJS, PostgreSQL, Prisma, Electron, Docker

- Designed and built a full-stack, multi-store POS platform covering checkout, inventory, purchasing, returns, staff permissions, reporting, tax exports, and accounting workflows.
- Engineered offline invoice issuance with local cryptographic signing and post-sync server-side chain verification; implemented RBAC, audit logging, automated tests, CI, and Docker builds.

MyProfit: Finance Assistant and Trading Bot | Co-developer 2025

Python, Telegram bot

- Built the finance engine for expense analysis, monthly reporting, savings recommendations, trading-budget allocation, and trade reporting; integrated the bot and trading components.

Bura Card Game | Solo Developer In development

Python, Pygame, pytest

- Built a tested trick-taking game with rules logic separated from the interface, computer opponents, and local two-player play.

EXPERIENCE

Riverside International School | Mathematics and ICT Teacher (Part-time) Sep 2025 - Present

- Teach IGCSE Mathematics and ICT to Year 9 and Year 10 students while completing a full-time degree.

LEADERSHIP & ACTIVITIES

Phystech Buddy, MIPT | Organizer

- Support incoming students, many of them international, through their first-year transition and organize community events.

MIPT Football Team, Player | Certified Swimming Lifeguard

SKILLS

Programming: Python, C++, TypeScript

Frameworks & Tools: React, NestJS, PostgreSQL, Prisma, Electron, Docker, Git, pytest

Mathematics & Research: Algorithms, combinatorial optimization, graph theory, probability, linear algebra, computational experiments

Languages: Arabic (native), English (fluent), Russian (intermediate)